

PHASE 4 PROJECT

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SXSW SENTIMENT ANALYSIS

Understanding Public Perception of Brands at SXSW via Twitter Data



Overview

Dataset : [judge-1377884607_tweet_product_company](#)

- Each Tweet includes: company, product category, sentiment label
- Goal: Understand customer perception and brand performance

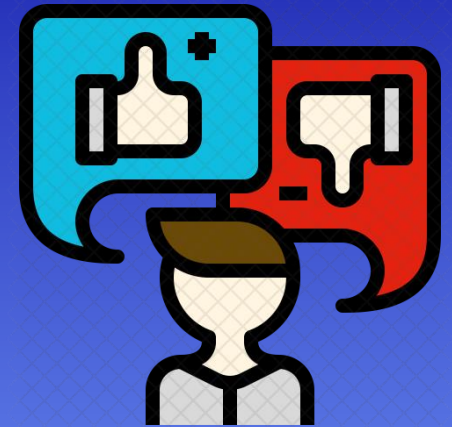


Data Overview

- Dataset source: Twitter API/ public tweets data
 - Timeframe: During SXSW event window
 - Brands: Apple&Google products and campaigns
 - Features: Tweet text,hashtags,timestamp,engagement metrics
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Business Understanding

- Companies need sentiment analysis to:
- Monitor public perception
- Detect dissatisfaction early
- Guide marketing and Product design
- Main Goal: To provide a data-driven view of Apple & Google products sentiments during SXSW.



Problem Statement

Key Questions:

- Is our campaign resonating positively?
- How do sentiments compare to competitors?
- What product features are driving sentiment?



Stakeholders

- Marketing Directors
- Brand Consultants
- Company Managers

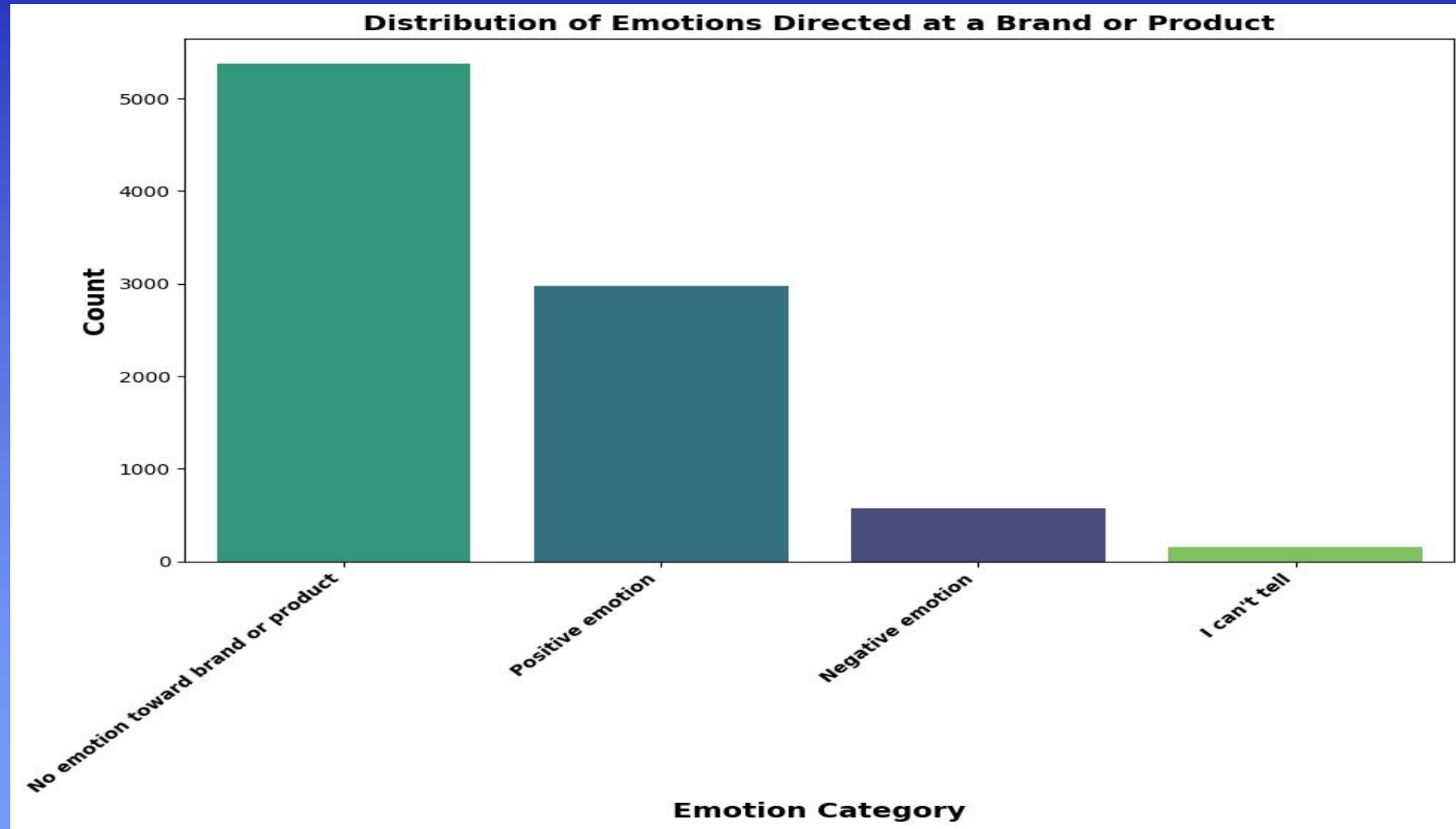


Analysis Approach

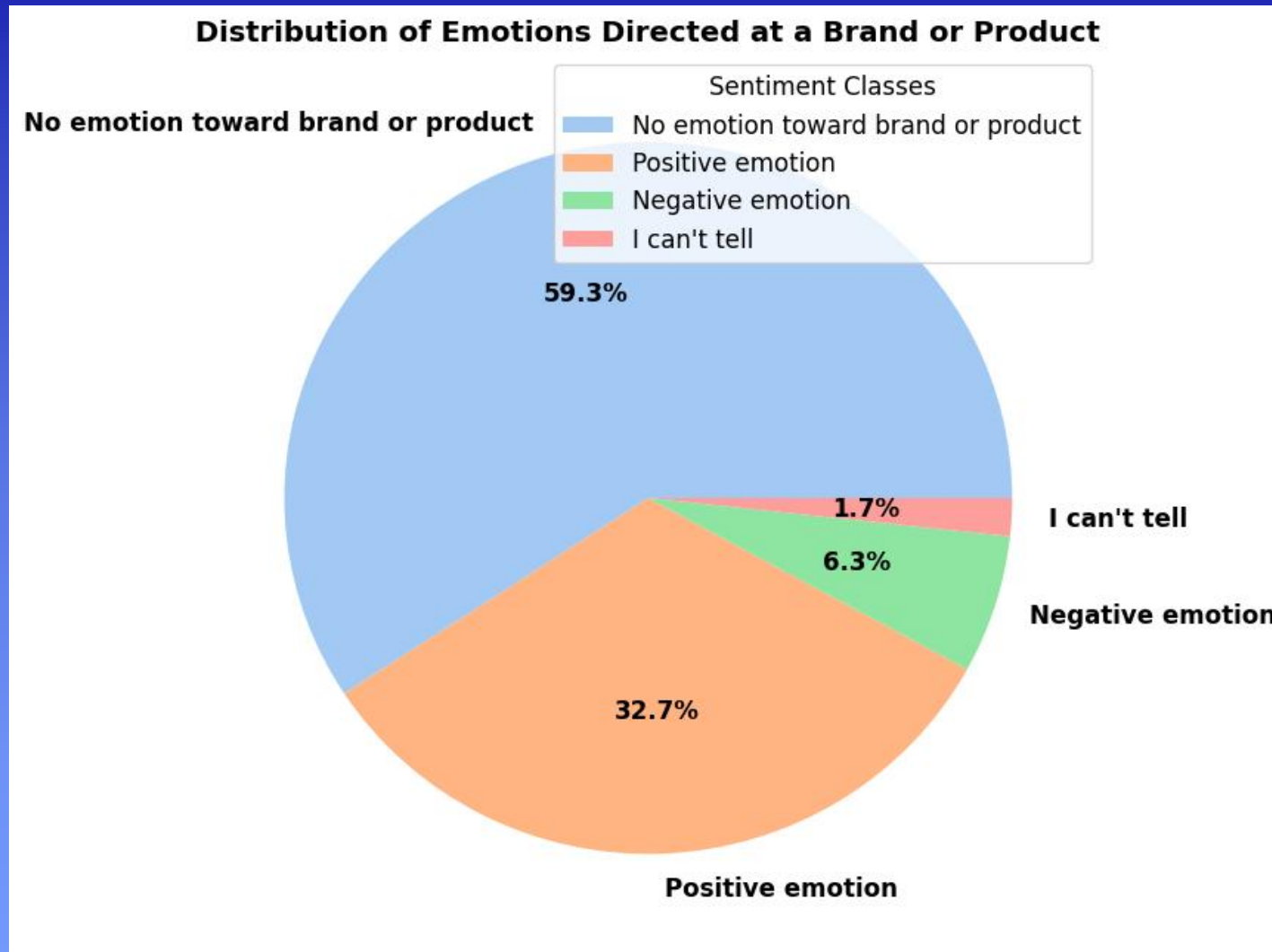
- Preprocessing: Cleaning,tokenization,encoding
- Model Analysis: ML& NLP model to classify tweets
- Topic Modeling: Identify emerging themes.
- Visualization: Trends and competitive heatmaps
- Insights



Emotion Category Distribution



Distribution of Emotions on Brand/Product



Modeling

- Models Tested:
- Logistic Regression(baseline)
- Random Forest(ensemble)
- Neural Network(non-linear)
- BERT(pre-trained transformer,best performer)

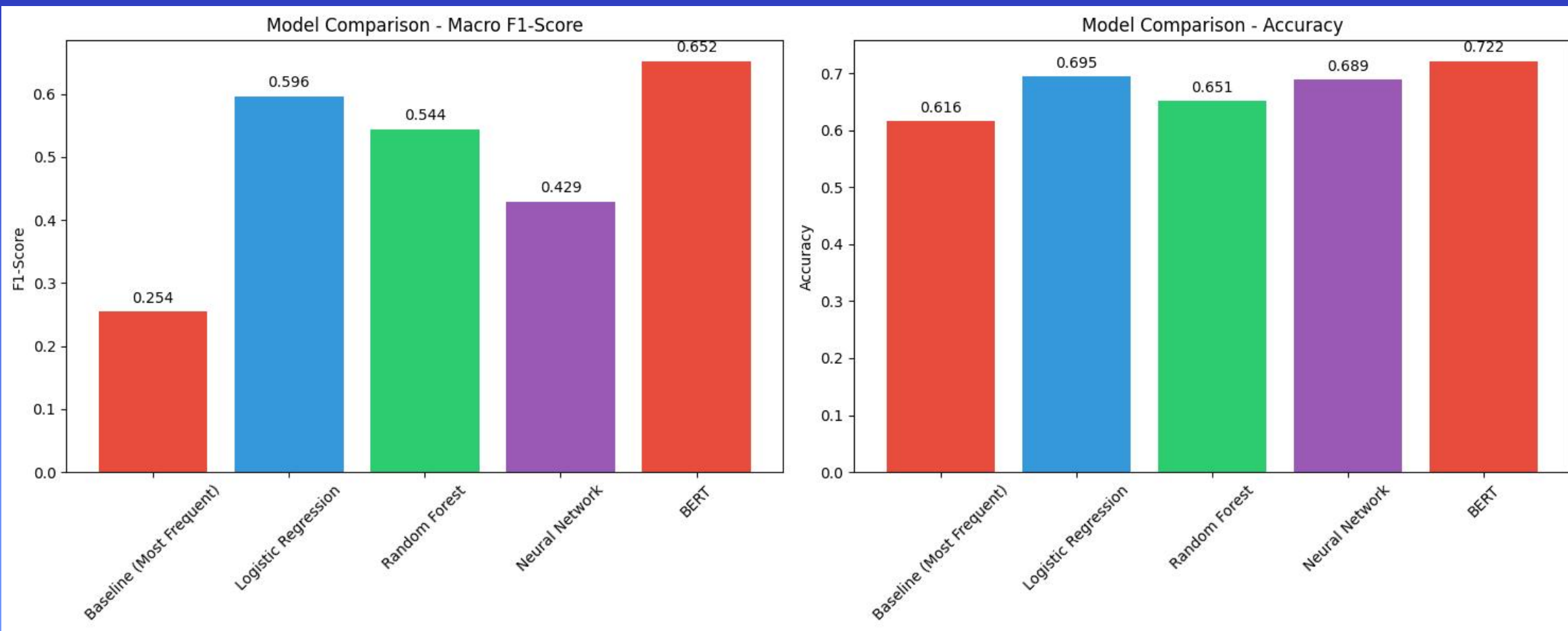


Results

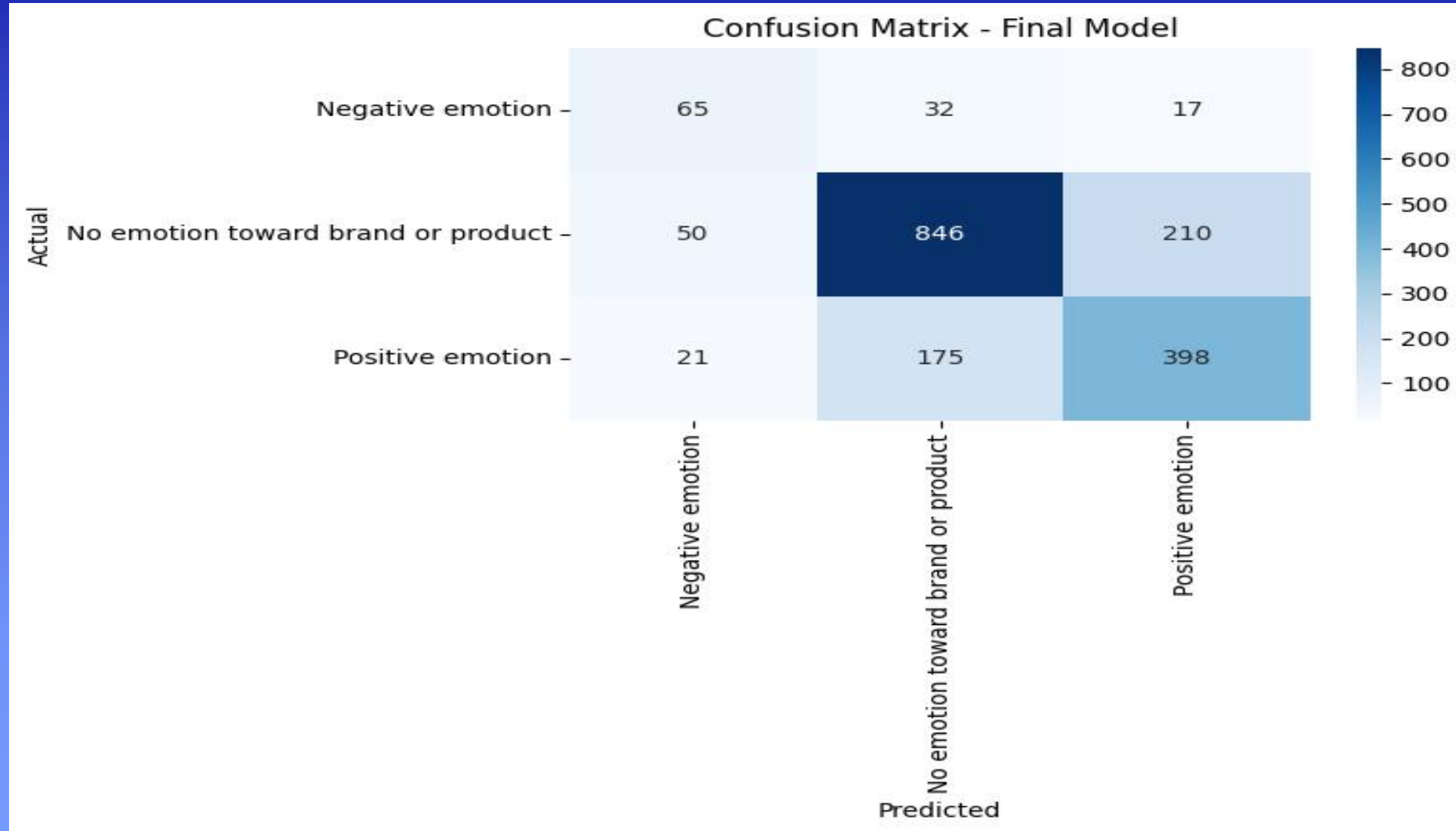
- Table;

MODEL	F1 SCORE	ACCURACY
Logistic Regression	0.59	0.69
Random Forest	0.54	0.65
Neural Network	0.31	0.67
BERT	0.62	0.72

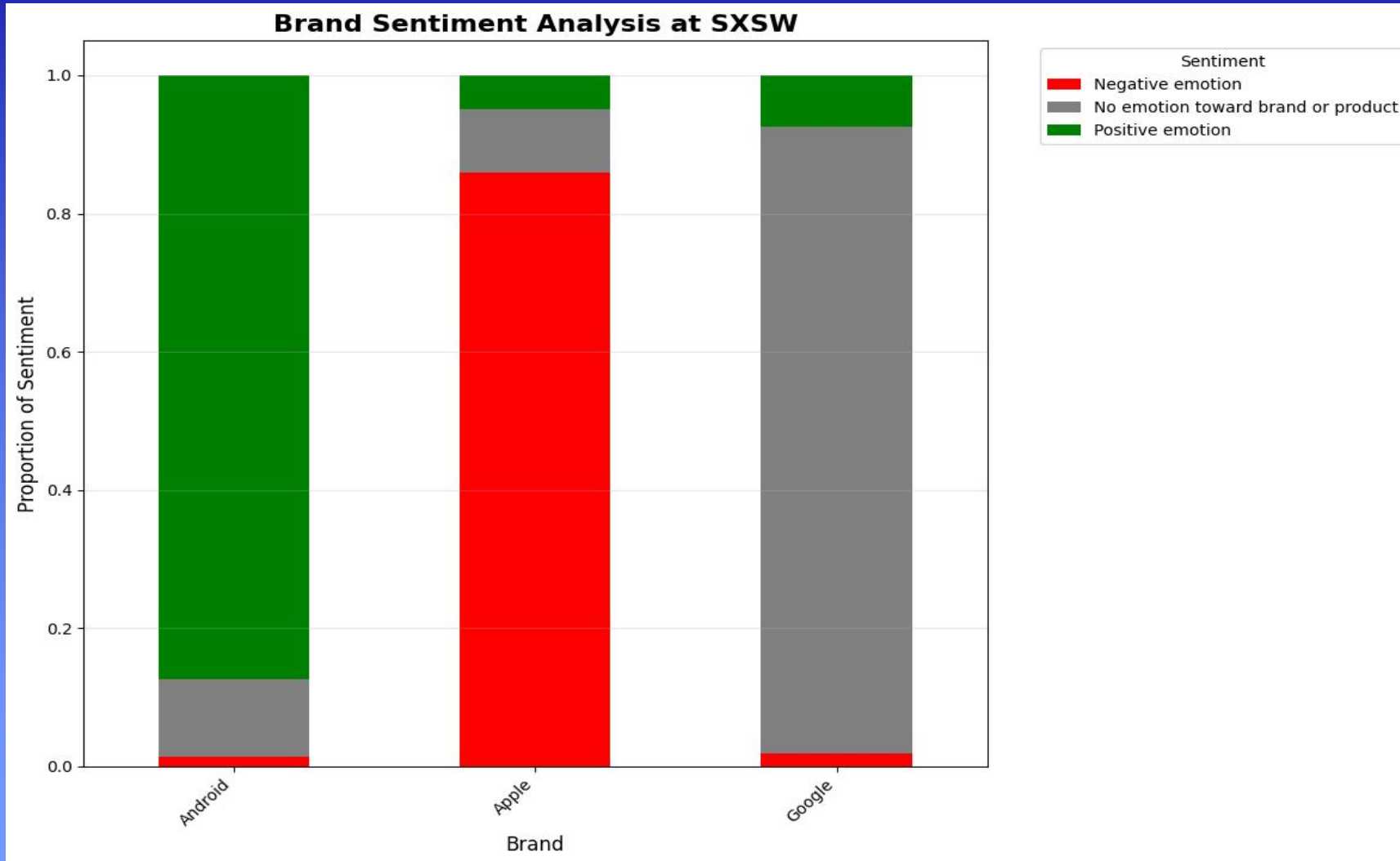
Model Comparison



Confusion Matrix



Brand Sentiment Analysis



Interpretation

- Apple faces criticism, Google remains neutral in the background while Android is winning on the positive tweets.



Key Insights

- Engagement strategies should focus on amplifying Android's strengths
- Address Apple's pain points
- Energizing Google's neutral feedback.
- Brands can leverage the positive sentiments for marketing amplification and campaigns.
- Neutral detection is strong, but bias toward neutrality.
- Negative sentiment under-detected.
- Competitive positioning visible in sentiment distribution.

Recommendations

To the Marketing Directors:

- Continue successful event strategies (Apple, Google)
- Monitor negative sentiments in real time



Continuation....

For Brand Consultants:

- Use sentiment score for competitive positioning
- Optimize live campaigns
- Detect emerging trends and market gaps



Continuation...

Finance Managers:

- Post-event analysis for following years's planning.
- Smarter budget allocation.



Business Value of this Analysis

- Data-Driven Decisions.
- Real- time competitive intelligence
- Return on Investment Measurement
- Proactive reputation Management

